

Two-Variable Rationalization and Sharp Stabilization of Cubic Threefolds

Tavis Rudd*

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Abstract

We construct a three-parameter family of smooth cubic threefolds whose least rational stabilization has dimension two over every extension of their characteristic-zero ground field. The construction rests on a surface theorem: every stably rational smooth quartic del Pezzo surface becomes rational after multiplication by the affine plane. A saturated rank-three subtorus of the projective Cox model has rational quotient, and equivariant torsor splitting leaves a rational two-dimensional torus. The cubic lower bound is supplied by the one-stabilization irrationality theorem. In an explicit rational pencil, we decompose intermediate Jacobians up to geometric isogeny into five elliptic factors and compute their potential toric ranks. These ranks separate a positive-density set of squarefree integral parameters. Hodge conservation then gives infinitely many pairwise nonbirational irrational fourfolds, each rational after one further projective-line factor. The rationalization and arithmetic calculations are independent of the two companion obstruction theorems.

Keywords. cubic threefold; quartic del Pezzo surface; stable rationality; universal torsor; variety with one apparent double point; algebraic torus; rationality after affine-line stabilization.

1 Introduction and main results

We ask for the least dimension of a projective-space factor that makes a stably rational variety rational. For a geometrically integral variety V/k , write

$$\ell_k(V) = \min\{m \geq 0 : V \times \mathbf{P}^m \text{ is } k\text{-rational}\},$$

with value infinity if the set is empty. Affine space gives the same definition. We determine this level on a three-parameter family of smooth cubic threefolds and distinguish infinitely many of their first stabilizations.

Let k have characteristic zero and put $B(x, y) = \lambda x^3 + \mu x^2 y + \nu y^3$, with $\lambda, \mu, \nu \in k$. Consider the cubic $\mathcal{X}_B \subset \mathbf{P}_k^4$ given by

$$(z - x)u^2 + 6yuv + 3(z + x)v^2 - z^3 + \frac{3}{4}(x^2 + 3y^2)z + B(x, y) = 0. \quad (1)$$

Theorem 1.1 (A family of exact level two). *If $\mathcal{X}_B$ is smooth, then for every field extension K/k ,*

$$\ell_K((\mathcal{X}_B)_K) = 2.$$

The family over $\mathbf{C}$ has a three-dimensional image in cubic moduli.

*tavisrudd@damnsimple.com

The upper bound comes from a quartic del Pezzo fibration. We construct a rational quotient of its projective Cox model by a rank-three torus; the remaining torus has dimension two and is rational. This proves the following surface theorem independently of any irrationality obstruction. A Galois lattice is *stably permutation* if it becomes a permutation lattice after adding a permutation lattice.

Theorem 1.2 (Two-dimensional stabilization). *Let k be a field of characteristic zero and let S/k be a smooth quartic del Pezzo surface. Suppose that $S(k) \neq \emptyset$ and that $\text{Pic}(S_{\bar{k}})$ is a stably permutation $\text{Gal}(\bar{k}/k)$ -module. Then*

$$S \times \mathbf{A}^2 \quad \text{and, equivalently,} \quad S \times \mathbf{P}^2$$

are k -rational. In particular, this holds for every stably k -rational smooth quartic del Pezzo surface.

The lower bound is the theorem that every smooth complex cubic threefold remains irrational after multiplication by $\mathbf{P}^1$, proved in *One-Stabilization Irrationality and Hodge Conservation for Fano Threefolds* [7, Theorem 1.1]. Finite descent transfers that obstruction to characteristic-zero fields. Thus the rationalization proved here attains the first possible level.

An arithmetic subfamily shows that the first stabilization retains many distinctions which the second erases. Write $\mathcal{X}_t$ for (1) with $B = ty^3$, and let $\mathcal{T}$ be the positive squarefree integers prime to six.

Theorem 1.3 (Separated first stabilizations). *For distinct $m, n \in \mathcal{T}$, the intermediate Jacobians $J(\mathcal{X}_m)$ and $J(\mathcal{X}_n)$ are not isogenous over $\mathbf{C}$, and $\mathcal{X}_m \times \mathbf{P}^1$ and $\mathcal{X}_n \times \mathbf{P}^1$ are not birational over $\mathbf{C}$. Every $\mathcal{X}_n \times \mathbf{P}^2$ is rational over $\mathbf{Q}$. Moreover,*

$$\#\{n \leq H : n \in \mathcal{T}\} = \frac{3}{\pi^2}H + O(\sqrt{H}).$$

The Jacobian calculation gives a local invariant with values zero, one and three: the dimension of the torus in eventual semistable reduction. A prime dividing exactly one of two squarefree parameters distinguishes their Jacobians. Rational Hodge conservation [7, Theorem 1.3] then distinguishes the first stabilizations. Consequently the smooth fourfolds $Y_n = \mathcal{X}_n \times \mathbf{P}^1$, $n \in \mathcal{T}$, are pairwise nonbirational and irrational even over $\mathbf{C}$, while every $Y_n \times \mathbf{P}^1$ is rational over $\mathbf{Q}$. These are birational assertions about products, not isomorphisms of affine cylinders.

The use of universal torsors in stable rationality begins with Beauville, Colliot-Thélène, Sansuc and Swinnerton-Dyer [2]. Tschinkel and Zhang supply the quartic del Pezzo Cox geometry, its one-apparent-double-point property, and the four minimal Galois types with stably permutation Picard lattice [17]. Their general stabilization bound is eleven; Theorem 1.2 lowers it to two. For their type I_0 , they attribute the bound two to Shepherd-Barron [15]. Their explicit cubic fibrations also give the named examples and higher-dimensional series in Section 4.

The Jacobian argument uses the conic-bundle isogeny theorem of Casalaina-Martin, Marquand and Zheng Zhang [8, Theorem 2.9]. Our family lies in the known non-Eckardt involution locus, and the pencil lies in Roulleau’s Klein-four locus [10, §3.3]. Explicit elliptic decompositions also have precedents [9, 11]; here the five factors give the local separation formula. The contribution and literature qualifications are recorded in the accompanying LITERATURE.md, rows N1–N3. We use geometric isogenies and do not assert a product principal polarization.

Let Z be the seven-dimensional projective Cox model attached to a rational point of S , let T_0 be its rank-five acting torus, and let $T_3 \subset T_0$ be the saturated rank-three subtorus we construct. The quotient has dimension $7 - 3 = 4 = 2 + (5 - 3)$ and two descriptions:

$$\underbrace{Z/T_3 \sim_k \mathbf{P}^4}_{\text{tangent section}}, \quad \underbrace{Z/T_3 \sim_k S \times (T_0/T_3)}_{\text{equivariant torsor splitting}}.$$

Four weight spaces with integral-basis differences make a codimension-three tangent section meet a general orbit once. Since the Cox model may be singular, we use the tangent-projection theorem in [17, Theorem 2.4], extending the smooth theorem of [3, Corollary 4.2].

Result	Construction here	Additional input
Two-variable upper bound	Sections 2–5	Cox geometry and tori
Exact cubic level	Section 5	One-step irrationality
Separated family	Section 6	Hodge conservation

Readers following the family theorem may proceed from the proof of Theorem 1.2 to Section 5, postponing the named examples in Section 4. Appendices A–D give optional consequences, finite-index extensions and the limits of the Cox-weight construction. Appendix E supplies the finite nonemptiness calculation used in Proposition 3.2; its coordinate details may be postponed on a first reading. Appendix F records the verification scope.

2 A criterion for rational torus quotients

The criterion below turns a linear section into a rational quotient. Choose $r + 1$ weight spaces whose weight differences form a basis of the character lattice. Signed maximal minors then give the unique translate of a general torus orbit that lies in a suitable codimension- r linear section. If the section contains the centre of a birational tangent projection, that projection parametrizes the relevant component of the section, and hence the quotient.

The descent hypothesis below is what turns this geometric slice over k^s into a k -rational quotient.

The integral-basis condition gives uniqueness of the orbit correction. For an effective action illustrating the distinction, let

$$t \cdot (x_0, x_1, x_2) = (x_0, t^d x_1, t x_2)$$

on $\mathbf{A}^3$, and impose $x_1 = x_0$. On $x_0 x_1 x_2 \neq 0$, the correcting elements satisfy $t^d = x_0/x_1$. In characteristic zero there are d geometric solutions; only for $d = 1$ is the correction unique. The action is effective because of the weight-one coordinate. This finite ambiguity is therefore different from a kernel in a redundant parametrization of the torus.

Let $Z \subset \mathbf{P}(V)$ be a geometrically integral n -fold over an infinite field k . At a smooth point $p \in Z(k)$, write

$$\pi_p : Z \dashrightarrow \mathbf{P}(V/\widehat{T}_p Z) \tag{2}$$

for projection from the embedded tangent space; the hat denotes the affine cone. In the application, $Z \subset \mathbf{P}^{2n+1}$ has one apparent double point: a general point of $\mathbf{P}^{2n+1}$ lies on a unique secant line to Z . Over an algebraically closed field, its tangent projection is birational in the smooth case by [3, Corollary 4.2]. The projective Cox model need not be smooth. The form used here is the stronger [17, Theorem 2.4], which applies to a reduced, irreducible, nondegenerate projective variety whose base change has one apparent double point and whose k -points are dense. The split Cox model is integral and its homogeneous ideal has no linear equations, so its twists satisfy the additional hypotheses.

Theorem 2.1 (A quotient criterion). *Let a rank- r k -torus T act projectively linearly and generically freely on $Z \subset \mathbf{P}(V)$, and suppose that the generic tangent projection π_p is birational onto $\mathbf{P}^n$. Choose a diagonal linear lift of the action over k^s . Its weights are determined up to a common translation. Suppose that it has nonzero weight spaces $V_{w_0}, \dots, V_{w_r}$ such that $w_1 - w_0, \dots, w_r - w_0$ is a $\mathbf{Z}$ -basis of $X^*(T_{k^s})$. Assume that the sum $W = V_{w_0} \oplus \dots \oplus V_{w_r}$ and the sum $\hat{B}$ of all other weight spaces descend to complementary k -subspaces of V ; write $B = \mathbf{P}(\hat{B})$.*

Suppose that there is a point $p \in Z(k)$ for which π_p is birational, together with a k -defined codimension- r linear subspace

$$\Lambda \supset \langle T_p Z, B \rangle$$

with equations $\lambda_1, \dots, \lambda_r$. Suppose there is an open $U_p \subset Z$ on which π_p is an isomorphism and a geometric point $x \in (\Lambda \cap U_p)(k^s)$ in the generically free locus such that the matrix

$$A(x)_{ij} = \lambda_i(x_j), \quad x_j \in V_{w_j}, \quad (3)$$

has rank r and all $r+1$ maximal minors nonzero. Then the Rosenlicht quotient Z/T , whose function field is $k(Z)^T$, is k -rational.

Proof. The maximal minors first produce a unique point of Λ on a general T -orbit. The hypothesis that this section meets U_p then identifies it birationally with projective space.

Orbit intersection and descent. For a geometric point $q = \sum q_j + q_B$, where $q_j \in V_{w_j}$ and $q_B \in \hat{B}$, put $A(q)_{ij} = \lambda_i(q_j)$ and define

$$\kappa_j(q) = (-1)^j \det A(q)_{\widehat{j}}, \quad 0 \leq j \leq r. \quad (4)$$

The equations of Λ vanish on B , so $tq \in \Lambda$ is equivalent to

$$A(q)(\chi^{w_0}(t), \dots, \chi^{w_r}(t))^t = 0.$$

The cofactor identity says that the kernel is generated by $(\kappa_0(q), \dots, \kappa_r(q))$. The basis hypothesis gives a unique $t(q) \in T(k^s)$ through

$$\chi^{w_j - w_0}(t(q)) = \frac{\kappa_j(q)}{\kappa_0(q)}, \quad 1 \leq j \leq r. \quad (5)$$

For $r = 1$, this says $\kappa_0 = \lambda_1(q_1)$, $\kappa_1 = -\lambda_1(q_0)$, and $\chi^{w_1 - w_0}(t) = \kappa_1/\kappa_0$: it is the unique scaling that makes $\lambda_1(tq) = 0$. Formula (5) is the same cofactor calculation in higher rank. Because the weight differences form a basis, the coordinates of $t(q)$ are Laurent monomials in the displayed ratios. The construction is unchanged if all weights are translated by one character, since only their differences occur. It is also independent of the ordering of the selected weight spaces: intrinsically, $s(q) = t(q)q$ is the unique point of $Tq \cap \Lambda$. Because W , $\hat{B}$, and Λ are defined over k , Galois conjugation preserves this incidence problem; uniqueness of the intersection point therefore makes s Galois equivariant. Thus s is a k -rational, orbit-invariant retraction on the open where the action is free and the minors do not vanish. Let C be the closure of its image. Then $C \dashrightarrow Z/T$ is birational and $C \subset \Lambda \cap Z$. Since $x \in \Lambda$, the vector $(1, \dots, 1)$ lies in the kernel of $A(x)$; uniqueness gives $s(x) = x$. Thus C meets U_p .

Rationality of the section. The image of Λ under π_p is a linear $\mathbf{P}^{n-r} \subset \mathbf{P}^n$, because Λ contains the centre of the linear projection. The irreducible variety C has dimension $n - r$. Since π_p is an isomorphism on U_p , its restriction identifies $C \cap U_p$ with a closed subset of the

projection's image open. This subset lies in the linear $\mathbf{P}^{n-r}$ and has the same dimension, so it is the intersection of that linear space with the image open. Thus

$$\pi_p|_C : C \dashrightarrow \mathbf{P}^{n-r} \quad (6)$$

identifies dense open subsets and is birational. Therefore Z/T is k -rational.

We record the inverse and its domain. Choose linear forms $\rho_0, \dots, \rho_{n-r}$ whose restrictions give homogeneous coordinates on the image of Λ in $\mathbf{P}(V/\widehat{T}_p Z)$. Let $U \subset Z$ be the invariant open where the action is free, all $\kappa_j(q)$ are nonzero, $s(q) \in U_p$, and $\rho_0(s(q)) \neq 0$. These conditions are invariant because s is orbit invariant and nonvanishing of the signed minors is invariant under the diagonal torus action. Then

$$\Phi : U/T \longrightarrow \mathbf{P}^{n-r}, \quad [q] \longmapsto [\rho_0(s(q)) : \dots : \rho_{n-r}(s(q))] \quad (7)$$

is the quotient parametrization. If $R_\alpha(x)$ are homogeneous equations for Z , the graph of its inverse over $\Phi(U/T)$ is the closure of the locus with $x \in C \cap U_p$ satisfying

$$\begin{aligned} R_\alpha(x) &= 0 \quad \text{for every } \alpha, \\ \lambda_i(x) &= 0 \quad 1 \leq i \leq r, \\ y_j \rho_0(x) - y_0 \rho_j(x) &= 0 \quad 1 \leq j \leq n-r. \end{aligned} \quad (8)$$

Write its unique point over y as $x(y)$ and set $\Theta(y) = [x(y)]$. The last line gives $\Phi(\Theta(y)) = y$. Conversely, $s(q) \in \Lambda \cap U_p$ and has the same tangent-projection coordinates as $x(\Phi([q]))$; injectivity of $\pi_p|_{U_p}$ gives

$$\Theta(\Phi([q])) = [s(q)] = [q] \quad \text{in } U/T.$$

Thus both composites are the identity on the stated dense opens. In the Cox application, the R_α in (8) are the twenty quadrics in [17, Section 3]. Equations (5) and (8) give the rational map and its inverse without an unrecorded finite cover. $\square$

The proof is constructive once equations for Z , a tangent point, and the linear forms λ_i are given. The signed minors in (4) compute the orbit correction, and (8) recovers the inverse parametrization by elimination. For an executable parametrization of $S \times \mathbf{P}^2$ over a specified number field one must also provide a descended pair p, x , its slice equations, a generic torsor trivialization, coordinates on the residual torus, and the composed forward and inverse maps with their nonvanishing conditions. The graph construction proves existence from these inputs; it does not supply expanded coordinate formulas or an elimination bound.

3 The rank-three subtorus of the projective Cox model

This section verifies the hypotheses of Theorem 2.1 for the projective Cox model. Four facts from Tschinkel and Zhang enter. The model is seven-dimensional, tangent projection from a general point is birational, and the scalar quotient of its rank-six Néron–Severi torus has rank five. In the minimal case, the stable-permutation hypothesis restricts the Galois image to four conjugacy types $I_0, \dots, I_3$ in $W(D_5)$. A rational point on the surface gives Zariski-dense rational points on the model and a generic splitting of the universal torsor. We construct the rank-three subtorus and tangent section for I_3 ; subgroup containment gives the other types after a change of marking.

Let T be the Néron–Severi torus, so $X^*(T_k) = \text{Pic}(S_k)$. The Galois-invariant cocharacter

$$\lambda_{-K} : \mathbf{G}_m \longrightarrow T, \quad D \longmapsto (-K_S, D), \quad (9)$$

acts by simultaneous scalar multiplication on the sixteen Cox generators. Write $T_0 = T/\lambda_{-K}(\mathbf{G}_m)$, a torus of rank five. Over $\bar{k}$, the projective Cox model

$$Z_{\bar{k}} = \text{Proj Cox}(S_{\bar{k}}) \subset \mathbf{P}_{\bar{k}}^{15}$$

is a seven-dimensional variety with one apparent double point [17, Lemma 3.2].

In the marking of [17, Sections 3–4], E_i , L_{ij} , and Q are the Cox generators for an exceptional curve, the line through two blowup points, and the conic through all five points, respectively. If H is the pullback of a line in $\mathbf{P}^2$, fix the character basis

$$b_i = E_i - E_5 \quad (1 \leq i \leq 4), \quad b_5 = H - 3E_5$$

of $X^*(T_{0,\bar{k}}) = (-K_S)^\perp \subset \text{Pic}(S_{\bar{k}})$, and write $e_1, \dots, e_5$ for the dual cocharacter basis. For the type- I_3 Galois image, the sublattice

$$N_3 = \mathbf{Z}e_3 \oplus \mathbf{Z}e_4 \oplus \mathbf{Z}e_5 \subset X_*(T_{0,\bar{k}}) \quad (10)$$

is saturated and Galois stable. The two displayed generators of the type- I_3 Galois image act on this basis by

$$\begin{bmatrix} -1 & 0 & 0 \\ -1 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & -1 \end{bmatrix}.$$

Let $T_3 \subset T_0$ be the corresponding rank-three subtorus.

Remark 3.1 (Saturation and the projective kernel). In the type- I_1 description, the three visible sign cocharacters in the last three coordinates are

$$(0, 1, 1), \quad (1, 0, 1), \quad (1, 1, 0).$$

They span the even-sum sublattice of $\mathbf{Z}^3$, of index two. Its saturation is $\mathbf{Z}^3$, obtained by adjoining

$$(1, 1, 1) = \frac{1}{2}((0, 1, 1) + (1, 0, 1) + (1, 1, 0)).$$

The map from the three parameter copies of $\mathbf{G}_m$ to their image in the projective torus has kernel μ_2 . The degree two obtained from the unsaturated parameters counts this kernel; it is not the degree of the Rosenlicht quotient. Saturation gives the integral-basis condition in Theorem 2.1.

We seek four projective weight spaces whose pairwise weight differences generate the full rank-three character lattice and whose complement is Galois stable. These conditions supply, respectively, the unique orbit correction and the descent required by Theorem 2.1. The following four groups of Cox generators span such weight spaces:

weight	Cox generators
$(0, 1, 1)$	L_{13}, L_{23}, L_{35}
$(1, 0, 1)$	L_{14}, L_{24}, L_{45}
$(1, 1, 0)$	E_1, E_2, E_5
$(1, 1, 1)$	L_{12}, L_{15}, L_{25}

These four weight spaces form a Galois-stable set, but they are not individually Galois stable. On their weights, the first displayed generator acts by

$$(011 \ 111)(101 \ 110),$$

and the second by (110 111), fixing 011 and 101. These are affine actions on projective weights: a different diagonal lift translates all four weights by a common character. Every other Cox generator lies in the Galois-stable subspace

$$B = \mathbf{P}\langle E_3, E_4, L_{34}, Q \rangle. \quad (11)$$

The selected simplex also identifies the acting torus intrinsically. Let Ω be its four weights and E_4/k the corresponding degree-four étale algebra. Common translations disappear in the map

$$\ker(\mathbf{Z}[\Omega] \xrightarrow{\Sigma} \mathbf{Z}) \longrightarrow X^*(T_3), \quad (n_w) \mapsto \sum_w n_w w.$$

The three differences are an integral basis, so this is a Galois-lattice isomorphism. Hence

$$T_3 \simeq \mathrm{Res}_{E_4/k} \mathbf{G}_m / \mathbf{G}_m.$$

Its underlying variety is the norm-nonzero open in $\mathbf{P}(E_4)$. This explains the simplex requirement; rationality of the quotient still requires the tangent-section argument.

By Galois stability, the direct sum of the four selected weight spaces and the affine cone over B are complementary subspaces defined over k , as required in Theorem 2.1. The determinant of the three weight differences from $(0, 1, 1)$ is one, supplying the integral-basis condition. On the open where the four selected weight-space components are nonzero, a projective stabilizer fixes their three ratios. The three weight differences form a basis, so that stabilizer is trivial. In particular, the T_3 -action is generically free.

Order the selected weights as 011, 101, 110, 111, and use splitting-field torus coordinates t_1, t_2, t_3 dual to e_3, e_4, e_5 . The signed minors κ_j of (4) give

$$\frac{t_1}{t_2} = \frac{\kappa_1}{\kappa_0}, \quad \frac{t_1}{t_3} = \frac{\kappa_2}{\kappa_0}, \quad t_1 = \frac{\kappa_3}{\kappa_0}.$$

Thus the orbit correction is explicitly

$$(t_1, t_2, t_3) = \left(\frac{\kappa_3}{\kappa_0}, \frac{\kappa_3}{\kappa_1}, \frac{\kappa_3}{\kappa_2} \right). \quad (12)$$

These are splitting-field coordinates. The corrected point descends by uniqueness of its intersection with the slice.

The residual character lattice has basis b_1, b_2 , and its two generators act on column coordinates by

$$A_{\mathrm{res}} = \begin{pmatrix} -1 & 0 \\ 1 & 1 \end{pmatrix}, \quad B_{\mathrm{res}} = \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix}.$$

They permute $(0, 1), (1, -1), (-1, 0)$; these generate the lattice with their sum as its only relation. Thus $T_0/T_3 \simeq \mathrm{Res}_{E_3/k}^1 \mathbf{G}_m$ for the cubic étale algebra of this Galois set. Both identifications restrict to the smaller Galois types.

The coordinate calculations below establish nonemptiness over the splitting field. The conditions themselves are defined over k ; density of k -points then supplies ground-field data. Descending those conditions does not require a particular displayed coordinate witness to descend.

Proposition 3.2 (A tangent section for the four Galois types). *Let k be a field of characteristic zero, let S/k be a smooth quartic del Pezzo surface, and let Z be its projective Cox model. Over k there exist a general tangent point p , a point x , and a codimension-three linear subspace $\Lambda \supset \langle T_p Z, B \rangle$ such that $x \in \Lambda \cap U_p$ and the maximal-minor condition of Theorem 2.1 holds at x for T_3 . If $S(k) \neq \emptyset$ and the Galois image has type I_0, I_1, I_2 , or I_3 , such a triple (p, x, Λ) exists over k .*

Proof. Work first over $\bar{k}$. For $p \in Z_{\text{sm}}$, let H_p be the space of tangent hyperplanes containing B . If $x = \sum_{j=0}^3 x_j + x_B$ is its weight-space decomposition, evaluation gives

$$\text{ev}_{p,x} : H_p \longrightarrow \bar{k}^4, \quad \lambda \longmapsto (\lambda(x_0), \dots, \lambda(x_3)).$$

Let $\mathcal{E}_{\text{ev}} \subset Z_{\text{sm}} \times Z$ be the open locus where H_p has dimension four, $\text{ev}_{p,x}$ is an isomorphism, and the T_3 -action is free at x . For the four Galois types in the statement, these conditions are Galois-invariant and therefore define a k -open. They are the conditions needed for the codimension-three orbit slice.

Geometric existence. First we show that $\mathcal{E}_{\text{ev}}$ is nonempty for every split Cox model attached to a smooth quartic del Pezzo surface. Use the standard blowup points

$$(1 : 0 : 0), (0 : 1 : 0), (0 : 0 : 1), (1 : 1 : 1), (1 : a : b), \quad \Delta = ab(a-1)(b-1)(a-b) \neq 0. \quad (13)$$

Appendix E records four exact split-coordinate witnesses and proves that their evaluation-and-smoothness opens cover $\{\Delta \neq 0\}$, as stated in (31). Hence $\mathcal{E}_{\text{ev}}$ is nonempty.

Compatibility with tangent projection. We impose the tangent-projection condition uniformly in p . Let $G \subset Z_{\text{sm}}$ be the nonempty open over which the tangent projection is birational. The affine tangent spaces form a subbundle $\widehat{\mathcal{T}} \subset V \otimes \mathcal{O}_G$. Put

$$\mathcal{Q} = (V \otimes \mathcal{O}_G) / \widehat{\mathcal{T}}.$$

The tangent projections are the fibres of the rational G -map

$$\Pi : G \times Z \dashrightarrow \mathbf{P}_G(\mathcal{Q}), \quad (p, x) \longmapsto (p, \pi_p(x)).$$

The induced map on the generic fibre of G is birational, so Π restricts to an isomorphism between dense open subsets of its source and target. Denote the source open by $\mathcal{U}$. Thus $(p, x) \in \mathcal{U}$ is a single open condition on the pair, and its fibre over p is an open U_p on which π_p is an isomorphism.

Since $G \times Z$ is geometrically integral, the intersection $\mathcal{U} \cap (\mathcal{E}_{\text{ev}}|_{G \times Z})$ is nonempty.

Choose (p, x) in this intersection. Then p is general for tangent projection, $x \in U_p$, the evaluation map is an isomorphism at x , and the T_3 -action is free at x .

The inverse image under $\text{ev}_{p,x}$ of the hyperplane $u_0 + u_1 + u_2 + u_3 = 0$ is the three-dimensional space of elements of H_p that vanish at x . Choose a basis $\lambda_1, \lambda_2, \lambda_3$ and let Λ be their common zero locus. Then $x \in \Lambda \cap U_p$, and the four maximal minors of the three evaluation rows are all nonzero. Thus the same point x satisfies both the minor condition and the tangent-projection condition.

Arithmetic descent. We now obtain p , x , and Λ over the ground field. Fix a k -form whose Galois image has one of the four types. Because T_3 , B , the set of four selected weight spaces, and the universal tangent-projection map are defined over k , the intersection $\mathcal{U} \cap (\mathcal{E}_{\text{ev}}|_{G \times Z})$ is a nonempty k -open in $G \times Z$. Its projection to G is dense constructible and therefore contains a nonempty open G_0 .

Corollary 3.5 of [17] supplies the universal torsor whose fibre over a fixed point of $S(k)$ is trivial. Its k -points are Zariski dense by [17, Lemma 2.1], and the same holds in the projective Cox model. Hence G_0 contains a point $p \in Z(k)$. The fibre $(\mathcal{U} \cap (\mathcal{E}_{\text{ev}}|_{G \times Z}))_p$ is a nonempty k -open in Z , so Zariski density supplies a point $x \in Z(k)$ in that fibre. The functional $H_p \rightarrow k$, $\lambda \mapsto \lambda(x)$, has a three-dimensional kernel, so a k -basis gives Λ . We do not claim that any one of the four displayed coordinate choices descends.

For type I_3 , stability of N_3 , B , and the set of four selected Cox weight spaces is checked on the two displayed generators. Up to conjugacy in $W(D_5)$, the groups of types I_0, I_1, I_2 are subgroups of the group of type I_3 [17, Lemma 4.2]. Restriction and a change of marking preserve the same data. The stable-permutation hypothesis implies that $H^1(H, \text{Pic}(S_{\bar{k}})) = 0$ for every subgroup H of the Galois image. In the minimal quartic del Pezzo case, Tschinkel and Zhang show that this leaves exactly the four types $I_0, \dots, I_3$ [17, Proposition 4.1 and Lemma 4.2]. This is not a classification of arbitrary subgroups of $W(D_5)$. $\square$

4 Proofs of the rationality statements

Proof of Theorem 1.2. We first treat the minimal case by taking a quotient of the projective Cox model. An equivariant generic trivialization then identifies that quotient with the product of S and a two-dimensional torus. The nonminimal case reduces to the classical rationality theorem in degree at least five.

Suppose first that S is k -minimal. Since a stably permutation module has vanishing H^1 on every open subgroup, Proposition 4.1 and Lemma 4.2 of [17] place the Galois image in one of the four types used in Proposition 3.2. Lemma 3.2 and Theorem 2.4 of [17], together with Theorem 2.1, give

$$Z/T_3 \sim_k \mathbf{P}^4. \quad (14)$$

By [17, Corollary 3.5], let $\mathcal{T} \rightarrow S$ be a universal torsor under the Néron–Severi torus T , chosen to have a k -point above a fixed point of $S(k)$. The stable-permutation identity for $X^*(T_{\bar{k}})$ implies that $T \times Q \simeq Q'$ for suitable quasi-trivial tori Q, Q' ; here quasi-trivial means that the character lattice is a permutation Galois module. Hilbert’s Theorem 90 therefore gives

$$H^1(F, T) = 0 \quad \text{for every field extension } F/k \quad (15)$$

by [17, Remark 2.2]. Thus the generic fibre over $k(S)$ has a point. Equivalently, the torsor has a rational section σ . Translation by σ defines the T -equivariant birational map

$$S \times T \dashrightarrow \mathcal{T}, \quad (s, t) \mapsto \sigma(s)t. \quad (16)$$

The split scalar subtorus in (9) acts as common multiplication on every Cox coordinate. The universal torsor is a dense open in the affine Cox cone, and its scalar quotient is a dense open in Z . Quotienting (16) by that scalar therefore gives a T_0 -equivariant birational map $S \times T_0 \dashrightarrow Z$. Since $T_3 \subset T_0$ is a k -subtorus, Rosenlicht quotients give

$$Z/T_3 \sim_k S \times (T_0/T_3). \quad (17)$$

The quotient T_0/T_3 is a two-dimensional k -torus, hence is k -rational by [19]. Equations (14) and (17) prove that $S \times \mathbf{A}^2$ is k -rational.

If S is not k -minimal, successively contracting k -defined unions of pairwise disjoint exceptional curves gives a del Pezzo surface S' of degree at least five. The image of a k -point of S is a k -point of S' , so S' , and therefore S , is k -rational by [18, Theorem 2.1].

Finally, if S is stably k -rational, a nonempty open of a projective space is birational to a nonempty open of $S \times \mathbf{P}^m$. Since k is infinite, this supplies a k -point on S . Manin’s stable birational Picard-module obstruction implies that $\text{Pic}(S_{\bar{k}})$ is stably permutation. In the form used here, this consequence of [6, Theorem 2.2] is recalled immediately before [17, Proposition 4.1] and included in [17, Corollary 4.3]. This proves the last assertion. $\square$

We first apply the surface theorem to the named examples and higher-dimensional series of Tschinkel and Zhang. Section 5 treats the three-parameter family of Theorem 1.1. Consider the smooth cubic threefolds

$$X_1 : (x_4 - 2x_3)x_1^2 + 3(x_4 + 2x_3)x_2^2 + 3x_3^2x_4 - x_4^3 + x_5^3 = 0, \quad (18)$$

$$X_3 : x_4(x_1^2 + 2x_1x_2) + x_3(x_1^2 + x_1x_2 + x_2^2) + x_3^3 - x_3x_4^2 + x_4^3 + 2x_5^3 = 0. \quad (19)$$

Theorem 4.1 (Exact level for the cubic threefolds). *For every field F of characteristic zero and $j \in \{1, 3\}$,*

$$\ell_F((X_j)_F) = 2.$$

Thus $(X_j)_F \times \mathbf{P}^1$ is not F -rational, whereas $(X_j)_F \times \mathbf{P}^2$ is F -rational.

For $r \geq 0$, let $X_{j,r} \subset \mathbf{P}_{\mathbf{Q}}^{r+4}$ be obtained from equation (18) or (19) by adding $w_1^3 + \cdots + w_r^3$ to its left-hand side. Thus $X_{j,0} = X_j$.

Corollary 4.2 (Two series of cubic hypersurfaces). *For every $r \geq 0$ and $j \in \{1, 3\}$, the variety $X_{j,r} \times \mathbf{P}^2$ is rational over $\mathbf{Q}$.*

Tschinkel and Zhang supply the smoothness, fibrations, and generic-fibre Galois types in [17, Propositions 5.1 and 5.3]. Applying Theorem 1.2 over the two base function fields proves Corollary 4.2 and sharpens their stable-rationality conclusions to $X_{j,r} \times \mathbf{P}^2$ rational.

Corollary 4.3 (Rationality after affine-line stabilization). *For $j \in \{1, 3\}$, the variety $Y_j = X_j \times \mathbf{P}^1$ is a smooth projective fourfold over $\mathbf{Q}$ which remains nonrational over $\mathbf{C}$, while $Y_j \times \mathbf{A}^1$ is rational over $\mathbf{Q}$.*

The fourfolds Y_j answer the question posed in [17, p. 13]: find a nonrational variety Y such that $Y \times \mathbf{A}^1$ is rational. This is a birational statement, not Zariski cancellation for isomorphic affine cylinders.

Corollary 4.4 (Variation in the moduli of cubic threefolds). *On the moduli space of smooth complex cubic threefolds, $\ell_{\mathbf{C}}((X_j)_{\mathbf{C}}) = 2$ for $j \in \{1, 3\}$, whereas $\ell_{\mathbf{C}}$ is infinite at a very general point.*

Proof of Corollary 4.2. The argument takes place entirely over the function field of the projection base. This avoids any specialization or fibrewise rationality claim.

For $X = X_{j,r}$, Tschinkel and Zhang project from a plane containing a distinguished line. The base is $\mathbf{P}_{\mathbf{Q}}^{r+1}$, with function field $K = \mathbf{Q}(t_1, \dots, t_{r+1})$, and the generic cubic-surface fibre is K -birational, after contracting the distinguished line, to a smooth quartic del Pezzo surface S/K with a K -point. Their calculation identifies the Galois image as type I_1 for $j = 1$ and type I_3 for $j = 3$ [17, Propositions 5.1 and 5.3]. Lemma 4.2 of [17] then shows that $\text{Pic}(S_{\bar{K}})$ is stably permutation. Theorem 1.2 gives

$$K(S)(u, v) \simeq K(z_1, z_2, z_3, z_4).$$

Since $\mathbf{Q}(X) = K(S)$ as extensions of K ,

$$\mathbf{Q}(X)(u, v) \simeq \mathbf{Q}(t_1, \dots, t_{r+1}, z_1, z_2, z_3, z_4)$$

is purely transcendental. This is rationality of $X \times \mathbf{A}^2$, equivalently of $X \times \mathbf{P}^2$. $\square$

Proof of Theorem 4.1. The one-stabilization theorem proves that $X \times \mathbf{P}^1$ is irrational for every smooth complex cubic threefold X [7, Theorem 1.1]. It applies to $(X_j)_{\mathbf{C}}$. Therefore $(X_j)_{\mathbf{C}}$ has no rational stabilization with $m \leq 1$. Suppose one existed over a characteristic-zero field F . A birational map to projective space, its inverse, and their composition identities use only finitely many coefficients. They therefore descend together to a finitely generated subfield $F_0/\mathbf{Q}$ of F , including the nonzero denominators defining their domains. An embedding $F_0 \hookrightarrow \mathbf{C}$ would give the forbidden rationalization over $\mathbf{C}$. No embedding of all of F is needed. Corollary 4.2 gives rationality at $m = 2$ over $\mathbf{Q}$, which base-changes to F . Hence the level is exactly two. $\square$

Proof of Corollary 4.3. Theorem 4.1 shows that $Y_j = X_j \times \mathbf{P}^1$ is nonrational, even after base change to $\mathbf{C}$. On the other hand,

$$\mathbf{Q}(Y_j \times \mathbf{A}^1) = \mathbf{Q}(X_j)(u, t)$$

is purely transcendental by Corollary 4.2. Equivalently, $\mathbf{P}^1 \times \mathbf{A}^1$ is birational to $\mathbf{A}^2$. Hence $Y_j \times \mathbf{A}^1$ is rational. $\square$

Proof of Corollary 4.4. Theorem 4.1 gives the asserted value for X_1 and X_3 . Engel, de Gaay Fortman, and Schreieder prove that a very general smooth complex cubic threefold does not admit a decomposition of the diagonal and hence is not stably rational [5, Corollary 1.4]. For such a threefold no finite m makes $X \times \mathbf{P}^m$ rational, so $\ell_{\mathbf{C}}(X) = \infty$. $\square$

5 The uniform cubic family

The surface theorem applies to every smooth member of (1). The point to check is the Galois action on the generic surface: special coefficients may shrink the splitting group, so we prove containment in a fixed signed action rather than assume generic monodromy persists.

Proof of Theorem 1.1. We contract the distinguished line in the cubic-surface fibre to obtain a quartic del Pezzo model. The following equations make the birational map explicit. Put $a = x/y$, $A = a^2 + 3$, $\beta = B(a, 1)$, and $F = k(a)$. Projection to $[x : y]$ has generic fibre birational to $S = \{Q_1 = Q_2 = 0\} \subset \mathbf{P}_F^4$, where

$$\begin{aligned} Q_1 &= -au^2 + 6uv + 3av^2 + \beta w^2 - zh, \\ Q_2 &= u^2 + 3v^2 + \frac{3}{4}Aw^2 - z^2 + wh. \end{aligned} \tag{20}$$

Indeed, $wQ_1 + zQ_2$ gives the cubic-surface equation, and on $w \neq 0$ the second quadric recovers h . The point $P = [0 : 0 : 0 : 0 : 1]$ is rational and smooth. With symmetric matrices for the quadrics, put $c(T) = T^3 - \frac{3}{4}AT - \beta$; then

$$\det(Q_1 + TQ_2) = \frac{3}{4}(T^2 - A)c(T), \quad \text{disc}(c) = \frac{27}{16}A^3 - 27\beta^2. \tag{21}$$

The rational function A^3 has odd valuation at the simple zeros of $a^2 + 3$, even after extending constants, and so is not a square. Thus the cubic discriminant is nonzero. If $s^2 = A$, then $c(s)c(-s) = \beta^2 - A^3/16 \neq 0$. The pencil has five distinct roots, so S is a smooth quartic del Pezzo surface.

A line on S through P would have direction in $w = z = 0$. The remaining equations give $u^2 + 3v^2 = 0$ and $-au^2 + 6uv + 3av^2 = 0$. A nonzero direction would force $u = -av$

and $a^2 + 3 = 0$, impossible in F . Thus P lies on no exceptional line. Its blowup is a smooth cubic surface, and projection from the distinguished line gives the conic bundle

$$(T - a)u^2 + 6uv + 3(T + a)v^2 = c(T)w^2. \quad (22)$$

The components of its five singular fibres encode the five pairs of ruling classes of S , as in [17, Section 4].

Let ρ_1, ρ_2, ρ_3 be the roots of c , with cyclic differences $d_1 = \rho_2 - \rho_3$, $d_2 = \rho_3 - \rho_1$, $d_3 = \rho_1 - \rho_2$. The coefficient relations give $d_i^2 = 3(A - \rho_i^2)$. At $T = \rho_i$ the component equations are

$$(\rho_i - a)u + (3 \pm d_i)v = 0.$$

They are nonzero: $c(a) = a^3/4 - 9a/4 - \beta$ cannot vanish identically, because $\beta = \lambda a^3 + \mu a^2 + \nu$ has no linear term. For the other two fibres choose square roots $e_+^2 = (s - a)c(s)$, $e_-^2 = (-s - a)c(-s)$. Their components are

$$(s - a)u + 3v = \pm e_+ w, \quad (-s - a)u + 3v = \pm e_- w.$$

The compatibility relation is

$$e_+^2 e_-^2 = \frac{\text{disc}(c)}{9} = \frac{(d_1 d_2 d_3)^2}{9}. \quad (23)$$

Choose signs so that $e_+ e_- = d_1 d_2 d_3 / 3$. An even root permutation permutes the d_i , while an odd permutation also negates all three. Relation (23) forces the corresponding parity of exchanges in the last two fibre pairs. These five pairs of singular-fibre components encode the del Pezzo ruling data. Label them by $\rho_1, \rho_2, \rho_3, s, -s$, and let c_i exchange the two components in pair i . The action is contained in

$$\langle (45), c_4 c_5, (123), c_1 c_2 c_3 c_5 (23) \rangle \subset W(D_5). \quad (24)$$

This is the signed Picard action identified as type I_3 at the end of [17, Proposition 5.3]. Its Picard lattice is stably permutation by [17, Lemma 4.2], and remains so on restriction to any subgroup. Theorem 1.2 therefore applies over F , giving

$$k(\mathcal{X}_B)(r_1, r_2) = F(S)(r_1, r_2) \simeq k(a, z_1, z_2, z_3, z_4).$$

This proves the upper bound over k and hence over every extension.

An alleged rationalization after at most one variable over K would descend, together with the cubic equation, its inverse and the composition identities, to a finitely generated subfield of characteristic zero. Embedding that subfield in $\mathbf{C}$ contradicts [7, Theorem 1.1]. Smoothness is preserved in this descent. Thus the exact level is two.

Finally consider the member $(\lambda, \mu, \nu) = (0, 0, 1)$. The exact Jacobian calculation shows it is smooth: the sixth power of each homogeneous coordinate lies in the ideal generated by its five partial derivatives. In the 35-dimensional space of cubic forms, the infinitesimal GL_5 -orbit has rank 25; adjoining the three coefficient directions $x^3, x^2 y, y^3$ raises the rank to 28. Hence the moduli map has rank three there, and its image has dimension three. The finite rank and ideal-membership checks are described in the verification appendix. $\square$

6 An arithmetic pencil with separated first stabilizations

The separation proof uses one output of the Jacobian calculation: a local rank formula. For $t = a/b \in \mathbf{Q}^*$ in lowest terms with $b > 0$, put $D_t = 16a^2 - 27b^2$. Let $\rho_p(t)$ be the dimension of the torus in the semistable reduction of $J(\mathcal{X}_t)$ after a sufficiently large finite extension of $\mathbf{Q}_p$. This is the *potential toric rank*. It is invariant under geometric isogeny.

Proposition 6.1 (Local rank formula). *Every $\mathcal{X}_t$, $t \in \mathbf{Q}^*$, is smooth. For $p \geq 5$,*

$$\rho_p(t) = \begin{cases} 1, & p \mid a, \\ 3, & p \mid D_t, \\ 0, & \text{otherwise.} \end{cases} \quad (25)$$

The two nonzero cases are disjoint; denominator primes have rank zero.

We prove the formula by reducing the intermediate Jacobian to five elliptic curves. No independence or pairwise nonisogeny of those factors is required.

6.1 The discriminant and its covers

The involution $(u, v) \mapsto (-u, -v)$ fixes the line $x = y = z = 0$ and the plane cubic $E_t : g = 0$, where

$$g = -z^3 + \frac{3}{4}(x^2 + 3y^2)z + ty^3.$$

The quadratic form in u, v has matrix

$$M = \begin{pmatrix} z - x & 3y \\ 3y & 3(z + x) \end{pmatrix},$$

whose determinant conic is $Q : z^2 = x^2 + 3y^2$. For $t \neq 0$, the substitution $[X : Y : Z] = [-12ty - 9z : 36tx : z]$ gives a Weierstrass equation for E_t :

$$Y^2 = X^3 - 243X + (1728t^2 - 1458), \quad \Delta(E_t) = -2^{12}3^9t^2(16t^2 - 27). \quad (26)$$

Parametrizing Q by $(x, y, z) = (r^2 - 3s^2, 2rs, r^2 + 3s^2)$ restricts $4g$ to $32tr^3s^3 - (r^2 + 3s^2)^3$, whose sextic discriminant is $2^{26}3^{12}t^4(16t^2 - 27)$. Thus E_t is smooth and meets Q transversely when $t(16t^2 - 27) \neq 0$.

These conditions also prove smoothness of $\mathcal{X}_t$. At a putative singular point with M invertible, the u, v derivatives force $u = v = 0$, and the point would be singular on E_t . If M has rank one and $(u, v) \neq 0$, its adjugate is proportional to $(u, v)(u, v)^t$; the remaining derivatives would force nontransversality of E_t and Q . The case $(u, v) = 0$ again would make E_t singular. If $M = 0$, then $x = y = z = 0$, and the three derivatives $-u^2 + 3v^2, 6uv, u^2 + 3v^2$ have no common projective zero. Conversely, each excluded parameter gives a singular point on the fixed plane cubic and hence on $\mathcal{X}_t$. In particular all nonzero rational parameters are smooth.

Projection of E_t to $[x : y]$ is a connected degree-three map with six simple branch values. Its Galois closure W has group S_3 and genus four. The alternating quotient has genus two and, geometrically, equation

$$C_t : \eta^2 = 16t^2 - (x^2 + 3)^3. \quad (27)$$

Write $f_2 : W \rightarrow E_t$ for the quotient by a transposition. This is also the component double cover over E_t in the conic bundle: if z is one root of $z^3 - \frac{3}{4}(x^2 + 3y^2)z - ty^3$, the difference of the other two roots has square $3(x^2 + 3y^2 - z^2)$, exactly the square class splitting the binary quadratic form. The component double cover over Q has geometric equation

$$H_t : \xi^2 = 32tx^3 - (x^2 + 3)^3. \quad (28)$$

The conic-bundle theorem [8, Theorem 2.9] supplies an isogeny

$$(J(W)/f_2^*J(E_t)) \times J(H_t) \longrightarrow J(\mathcal{X}_t).$$

For the S_3 -action on $J(W)$, the invariant part has dimension zero, the sign part dimension two, and the standard part dimension two. Here dimensions are dimensions of abelian varieties. The standard rational representation has dimension two, with one-dimensional transposition invariants. Its elliptic multiplicity factor is therefore E_t , occurring twice in the standard part. Thus

$$J(W) \sim J(C_t) \times E_t^2, \quad J(\mathcal{X}_t) \sim E_t \times J(C_t) \times J(H_t). \quad (29)$$

These are geometric isogenies. Over $\mathbf{Q}$, the displayed genus-two equations require twists by -3 and -6 , respectively: the cubic discriminant is square-equivalent to $-3C_t$; on $\mathcal{Q}$ the quadratic form is $6(su + rv)^2$, giving component square class $-H_t/24$. Twists disappear after finite extension, so they do not affect potential ranks. They must be retained for arithmetic point counts.

6.2 Elliptic quotients and reduction

The maps $(x, \eta) \mapsto (x^2, \eta)$ and $(x, \eta) \mapsto (x^2, x\eta)$ give elliptic quotients

$$E_0 : Y^2 = 16t^2 - (U + 3)^3, \quad E_2 : Y^2 = U(16t^2 - (U + 3)^3).$$

Their differential pullbacks are independent, whence $J(C_t) \sim E_0 \times E_2$; put $E_1 = E_t$. Set $s = 4t/(3\sqrt{3})$. After geometric scaling, H_t has equation $y^2 = (x^2 + 1)^3 - 8sx^3$. Its two elliptic quotient maps are

$$U = x + x^{-1}, \quad V_{\pm} = y(x \pm 1)/x^2, \quad E_{\pm} : V_{\pm}^2 = (U \pm 2)(U^3 - 8s).$$

Their differential pullbacks are again independent. Thus

$$J(\mathcal{X}_t) \sim E_0 \times E_1 \times E_2 \times E_+ \times E_-. \quad (30)$$

The j -invariants, obtained from these equations, are

E_0	E_1	E_2	E_+	E_-
0	$\frac{432}{s^2(1-s^2)}$	$\frac{-6912s^2}{(s^2-1)^2}$	$\frac{6912s}{(s+1)^2}$	$\frac{-6912s}{(s-1)^2}$

Writing $c = j(E_2)$, the last two invariants have sum $4c$ and product $6912c$.

Proof of Proposition 6.1. Smoothness was proved above. Potential toric rank is preserved by isogeny: an isogeny and a quasi-inverse extend to Néron models and induce maps on their torus parts whose composites are multiplication by a nonzero integer. The torus dimensions are therefore equal. After semistable reduction, they are unchanged by further finite extension; see [12, Corollary 3.3]. An elliptic curve is potentially good precisely when its j -invariant is integral [13], and otherwise has potential toric rank one.

In terms of $t = a/b$,

$$j(E_1) = -\frac{19683b^4}{a^2D_t}, \quad c = -\frac{2985984a^2b^2}{D_t^2}.$$

At $p \geq 5$, if c is integral, both roots of $Z^2 - 4cZ + 6912c$ are integral. If $v_p(c) < 0$, its Newton polygon gives root valuations $v_p(c)$ and zero. Since $j(E_0) = 0$, (30) gives

$$\rho_p(t) = \mathbf{1}_{v_p(j(E_1)) < 0} + 2\mathbf{1}_{v_p(c) < 0}.$$

If $p \mid a$, then D_t is a unit and only the first term contributes. If $p \mid D_t$, both a, b are units and both terms contribute. If $p \nmid b$, or none of a, b, D_t is divisible by p , both invariants are integral. This proves (25). Any complex isogeny between the Jacobians, which are defined over $\mathbf{Q}$, descends to a finite number-field extension. Choosing a place over p therefore shows that different potential ranks rule out isogeny even over $\mathbf{C}$. $\square$

Proof of Theorem 1.3. For distinct $m, n \in \mathcal{T}$, some prime $p \geq 5$ divides exactly one, say m . Then $\rho_p(m) = 1$ but $\rho_p(n) \in \{0, 3\}$. Proposition 6.1 therefore rules out geometric isogeny. If the first stabilizations were birational, rational Hodge conservation [7, Theorem 1.3] would identify their rational third cohomology, and hence their intermediate Jacobians up to isogeny, a contradiction. The upper bound is Theorem 1.1.

For the count, use $\mathbf{1}_{\text{squarefree}}(n) = \sum_{d^2 \mid n} \mu(d)$. Restricting to integers prime to six gives

$$\#\{n \leq H : n \in \mathcal{T}\} = \frac{H}{3} \sum_{(d,6)=1} \frac{\mu(d)}{d^2} + O(\sqrt{H}) = \frac{3H}{\pi^2} + O(\sqrt{H}).$$

Truncation at $d \leq \sqrt{H}$ contributes the stated error, as does counting residue classes prime to six. This count uses no binary-form sieve. $\square$

A Generic surfaces, partners, and fibrations

The same function-field argument shows that the surface bound itself is sharp. Let $S_j/\mathbf{Q}(t)$ be the generic quartic del Pezzo surface obtained from X_j in the proof of Corollary 4.2.

Corollary A.1 (Exact level of the generic surfaces). *For every field F of characteristic zero and $j \in \{1, 3\}$,*

$$\ell_{F(t)}((S_j)_{F(t)}) = 2.$$

Proof. The rationalization of $S_j \times \mathbf{A}^2$ over $\mathbf{Q}(t)$ constructed in the proof of Corollary 4.2 base-changes to $F(t)$. The same generic-fibre identification gives $F(X_j) = F(t)((S_j)_{F(t)})$. If the surface were rational after one stabilization, then

$$F(X_j)(u) \simeq F(t)(z_1, z_2, z_3)$$

would be purely transcendental over F , contradicting Theorem 4.1. Rationality without stabilization would also imply rationality after one stabilization. This proves the lower bound. The argument does not require the Galois image to retain its original type after extension of the ground field. $\square$

Exact level also transfers through the projective-line correspondences used in the companion paper.

Corollary A.2 (Partners of the specified cubics). *Let F have characteristic zero and let V/F be a geometrically integral threefold. If, for some $j \in \{1, 3\}$,*

$$V \times \mathbf{P}^1 \sim_F (X_j)_F \times \mathbf{P}^1,$$

then $\ell_F(V) = 2$.

Proof. The displayed product is irrational by Theorem 4.1. Taking one further product with $\mathbf{P}^1$, and using $\mathbf{P}^1 \times \mathbf{P}^1 \sim_F \mathbf{P}^2$, gives

$$V \times \mathbf{P}^2 \sim_F (X_j)_F \times \mathbf{P}^2 \sim_F \mathbf{P}^5.$$

Thus two stabilizations suffice and one does not. $\square$

Remark A.3 (Genus-eight Fano partners). Over an algebraically closed field of characteristic zero, Kuznetsov's flop relates rank-two projective bundles over a smooth genus-eight Fano threefold and its associated Pfaffian cubic [16, Theorems 2.17–2.18]. A projective bundle is birational to the product with projective space. Hence Corollary A.2 applies when the associated cubic is $(X_1)_F$ or $(X_3)_F$. Over other fields the same conclusion holds whenever that correspondence is defined over F . It is not an exact-level statement for arbitrary genus-eight Fano threefolds.

For a general fibration the base contributes its own stabilization level. The following formula keeps track of the four free variables supplied by the surface rationalization.

Proposition A.4 (Stabilization along a surface fibration). *Let k have characteristic zero, and let $Y \dashrightarrow B$ be a dominant rational map of geometrically integral k -varieties. Suppose its generic fibre is $k(B)$ -birational to a smooth quartic del Pezzo surface satisfying the hypotheses of Theorem 1.2. Then*

$$Y \times \mathbf{A}^2 \sim_k B \times \mathbf{A}^4, \quad \max\{\ell_k(Y) - 2, 0\} = \max\{\ell_k(B) - 4, 0\},$$

where subtracting a finite integer from ∞ leaves ∞ . In particular, $\ell_k(B) \leq 4$ implies $\ell_k(Y) \leq 2$; if $4 < \ell_k(B) < \infty$, then $\ell_k(Y) = \ell_k(B) - 2$. The variety Y is stably rational if and only if B is.

Proof. Put $K = k(B)$. The surface theorem gives

$$k(Y)(u, v) \simeq K(z_1, z_2, z_3, z_4),$$

which proves the birational equivalence over k . For every geometrically integral variety W and integer $a \geq 0$, the definition of the level gives

$$\ell_k(W \times \mathbf{A}^a) = \max\{\ell_k(W) - a, 0\}.$$

Indeed, further stabilization by m makes this product rational exactly when stabilization of W by $a + m$ does; rationality persists under additional affine factors. This also proves the formula when the level is infinite. Apply it to the two birational varieties above. The stated consequences follow by considering whether $\ell_k(B)$ is at most four, finite and greater than four, or infinite. $\square$

Corollary A.5 (Exact trivial-stabilization linearization levels). *Over a characteristic-zero field k , each cubic $(X_j)_k$ gives a rational $\mathbf{G}_m^2$ -action on $\mathbf{A}_k^5$ which becomes birationally linear after exactly two added variables fixed by the torus. Every rank-one subtorus restriction requires exactly one such variable. Over $\mathbf{C}$, every finite-subgroup quotient is rational.*

Proof. Put $K = k(X_j)$ and identify $K(u, v)$ with a rational function field in five variables. Let the torus fix K and scale u, v . Its invariant field after m trivial variables is $K(s_1, \dots, s_m)$. A linear split-torus action over k has rational invariant field: diagonalize the action and choose an integral basis of the kernel of its weight map. The corresponding invariant monomials freely generate that field; compare [20, Theorem 4 and Lemma 5] over an algebraically closed field. Theorem 4.1 forces $m \geq 2$; at $m = 2$ rationalize the invariant field and retain u, v as diagonal coordinates. For a primitive rank-one subtorus of weights (a, b) , the invariant field is $K(u^b v^{-a})$, of exact stabilization level one. A complementary unimodular monomial is a diagonal coordinate, giving the same converse. For a finite subgroup over $\mathbf{C}$, the invariant monomial lattice has rank two, so its field is $K(r_1, r_2)$, which is rational. $\square$

Nonlinearizable rank-two rational torus actions on $\mathbf{A}^5$ already occur in Popov's Theorem 5 and Corollary 5; the assertion here specifies their exact trivial-stabilization levels for these cubic constructions. These are rational actions, not assertions about regular polynomial actions.

The same construction applied to $\mathcal{X}_n$, $n \in \mathcal{T}$, gives infinitely many rank-one rational actions on $\mathbf{A}_{\mathbf{Q}}^5$ which are pairwise nonconjugate even over $\mathbf{C}$: their invariant fields are the function fields of the pairwise nonbirational fourfolds $\mathcal{X}_n \times \mathbf{P}^1$. After one invariant variable is adjoined, each action is conjugate to the same diagonal action on $\mathbf{A}^6$.

B Finite rational partner sets in the pencil

The squarefree parameters give explicit distinct classes. On the entire rational pencil, the same local formula gives a different conclusion: each geometric isogeny class has only finitely many parameters.

Corollary B.1 (Finiteness within the rational pencil). *Fix $t \in \mathbf{Q}^*$. There are only finitely many $u \in \mathbf{Q}^*$ such that $J(\mathcal{X}_u)$ is geometrically isogenous to $J(\mathcal{X}_t)$. In particular only finitely many rational parameters u have $\mathcal{X}_u \times \mathbf{P}^1$ birational over $\mathbf{C}$ to $\mathcal{X}_t \times \mathbf{P}^1$.*

Proof. Let S be the finite set consisting of 2, 3 and the primes of positive potential toric rank for $J(\mathcal{X}_t)$. Every isogeny partner is potentially good outside S . Put $L = \mathbf{Q}(\sqrt{3})$ and let S_L be the finite places above S . For such a parameter u , set

$$w = \frac{3\sqrt{3}}{4u}, \quad x = \frac{1+w}{2}, \quad y = \frac{1-w}{2}.$$

Outside S , formula (25) gives $v(u) \leq 0$. Thus w is integral. If $v(w) > 0$, both $1+w$ and $1-w$ are units. If $v(w) = 0$, their product $(16u^2 - 27)/(16u^2)$ is a unit by the same rank formula. Consequently

$$x + y = 1, \quad x, y \in \mathcal{O}_{L, S_L}^*, \quad \sigma(x) = y, \quad x \neq y,$$

where σ is the nontrivial automorphism of $L/\mathbf{Q}$. The unit-equation theorem [14, Theorem 1.1] gives only finitely many such pairs. The inverse formula $u = 3\sqrt{3}/(4(x-y))$ makes the parameter map injective. The group of conjugate pairs has rank $1 + |S_L|$, giving the coarse bound $2^{8(|S_L|+2)}$ for this candidate set. Hodge conservation [7, Theorem 1.3] gives the last assertion. $\square$

The finite set contains necessary candidates, not a classification of the partners. No implemented unit-equation solver or affirmative birationality test is required for the corollary.

C Finite-index slices and zero-cycles

The tangent projection still supplies a rational component when the weight differences have finite index. The issue is to show that this particular component dominates the quotient and to determine its degree.

Theorem C.1 (Finite-index torus slices). *Assume that k has characteristic zero. Keep all the hypotheses of Theorem 2.1, except that*

$$M' = \sum_{j=1}^r \mathbf{Z}(w_j - w_0) \subset M = X^*(T_{k^s})$$

is required to have finite index d , rather than index one. The closure C of $\Lambda \cap U_p$ is a geometrically integral, k -rational variety, and its rational map to Z/T is dominant and generically finite of degree e dividing d . In particular, the quotient admits a dominant rational parametrization $\mathbf{P}^{n-r} \dashrightarrow Z/T$ of degree $e \mid d$.

Proof. The image L of Λ under tangent projection is a k -linear $\mathbf{P}^{n-r}$. Take the isomorphism open U_p to be defined over k . Since Λ contains the projection centre,

$$\Lambda \cap U_p \simeq L \cap \pi_p(U_p).$$

The right side is a nonempty open in L , by the hypothesis on x . Thus its closure C is defined over k , geometrically integral and k -rational. This constructs the component without choosing a geometric component and then assuming that it descends.

We next prove dominance. The selected weight spaces are permuted by Galois, up to a common translation of weights, so M' is Galois stable. Its inclusion in M defines an isogeny of k -tori

$$\varphi : T \longrightarrow T', \quad X^*(T'_{k^s}) = M',$$

with finite étale kernel H of order d . Over k^s , its coordinates are $\chi^{w_j - w_0}$, for $1 \leq j \leq r$. At the chosen point $x \in \Lambda \cap U_p$, the equations of Λ , restricted to the orbit, have invertible differential in the torus directions. Indeed, $A(x)(1, \dots, 1)^t = 0$; after removing the common character, their differential is the product of the last r columns of $A(x)$ and $d\varphi$. The first matrix is invertible because $\kappa_0(x) \neq 0$, and $d\varphi$ is invertible in characteristic zero. Therefore the orbit tangent space and $T_x C$ are complementary in $T_x Z$. The action map $T \times C \rightarrow Z$ is étale at $(1, x)$, hence dominant. Consequently C meets an invariant quotient open and dominates Z/T . Its dimension is $n - r$, so the induced map is generically finite.

To prove divisibility of its degree, extend the constants to $\bar{k}$ and put $F = \bar{k}(Z/T)$. The generic orbit is a torsor under the split torus T_F ; Hilbert's Theorem 90 gives an F -point on it. Using that point to identify the orbit with T_F , its intersection with Λ is given by (5). It is an H_F -torsor of degree d . Here H is constant, since the constants are algebraically closed. The absolute Galois group of F acts on this torsor by translations through a subgroup of H . Every orbit therefore has the order of that subgroup, a divisor of d . Equivalently, every field factor of this finite étale algebra has degree dividing d .

The generic fibre of $C_{\bar{k}} \rightarrow (Z/T)_{\bar{k}}$ is one such field factor. Indeed, $C \cap U_p$ is an open in the full linear section, and C is geometrically integral. Its degree is still e : a generically finite morphism of degree e becomes finite locally free of rank e after shrinking the target, and that rank is preserved by extension of constants. Thus $e \mid d$. $\square$

Remark C.2 (The degree can be smaller than the index). Let $Z = \mathbf{P}^1 \times \mathbf{P}^1 \subset \mathbf{P}^5$ be embedded by $\mathcal{O}(1, 2)$. On an affine chart its coordinates are

$$[1 : x : y : xy : y^2 : xy^2].$$

Let $\mathbf{G}_m$ scale y , giving weights $0, 1, 2$, each with multiplicity two. Select the weight-zero and weight-two spaces, so $d = 2$, and take $p = (1, 1)$. The hyperplane

$$\Lambda : X_5 - X_4 - X_1 + X_0 = 0$$

contains the tangent plane at p and the weight-one space. Its equation on Z is $(x - 1)(y^2 - 1) = 0$. In coordinates $u = x - 1$, $v = y - 1$, tangent projection is

$$(u, v) \longmapsto [uv : v^2 : uv^2],$$

with inverse $u = Y_2/Y_1$, $v = Y_2/Y_0$ on the open where all three target coordinates are nonzero. Thus it is an isomorphism on $uv \neq 0$. Automorphisms of the two factors give the same conclusion at every tangent point. The point $(x, y) = (2, -1)$ lies in this open and in Λ ; the two entries of $A(x, y)$ are $-1, 1$. The component meeting the open is $y = -1$. It maps birationally to the quotient, whose function field is $k(x)$, so $e = 1 < d = 2$. The other horizontal component, $y = 1$, and the vertical component $x = 1$ lie outside the isomorphism open. This example satisfies the tangent and descent hypotheses; it is not merely a weight calculation.

Corollary C.3 (Zero-cycles from finite-index slices). *Let Q be a geometrically integral variety over a characteristic-zero field k . Suppose that it is k -birational to quotients admitting constructions as in Theorem C.1, with indices $d_1, \dots, d_s$, where $s \geq 1$. For every smooth proper model $\overline{Q}$ and every extension F/k , the degree map*

$$\deg : CH_0(\overline{Q}_F) \longrightarrow \mathbf{Z}$$

is surjective and its kernel is annihilated by $\gcd(d_1, \dots, d_s)$. In particular, two coprime indices imply universal CH_0 -triviality.

Proof. The theorem gives parametrizations of degrees $e_i \mid d_i$. A parametrization has a nonempty open domain in projective space. Since k is infinite, this domain has a k -point, whose image supplies a point of $\overline{Q}(k)$. Its base changes make all degree maps surjective.

For clarity, recall the degree argument of [4, Lemma 6.2 and Propositions 6.3–6.4]; the smooth proper form is also stated in [1, Section 1.2]. Resolve one parametrization to a proper generically finite morphism $f : W \rightarrow \overline{Q}$ of degree e_i , with W smooth, projective and k -rational. Over any F/k , birational invariance gives $CH_0(W_F) \simeq \mathbf{Z}$ by degree. Shrink to a nonempty open of $\overline{Q}_F$ over which f_F is finite flat of degree e_i . The moving lemma represents any degree-zero class α by a cycle supported in that open. Its flat inverse image is a degree-zero cycle on W_F , hence is rationally equivalent to zero. Pushing it forward gives $e_i \alpha = 0$. Thus also $d_i \alpha = 0$ for each i ; a Bézout combination proves the assertion. $\square$

The same integer $D = \gcd(d_1, \dots, d_s)$ bounds diagonal torsion. For a smooth proper model $\overline{Q}$, choose the rational point q supplied in the proof and apply the conclusion over $k(\overline{Q})$ to the generic point minus q . Chow localization gives

$$D[\Delta_{\overline{Q}}] = D[\overline{Q} \times q] + Z, \quad \text{supp } Z \subset D' \times \overline{Q}, \quad D' \subsetneq \overline{Q}.$$

Thus its torsion order divides D , with the two factors transposed relative to [21, Definition 1.1 and Lemma 1.3]. It need not equal D .

The coprime-index conclusion is a criterion for future constructions. The scroll above illustrates the degree calculation on a rational quotient; it does not supply a new nonrational example with coprime-index slices.

D Limits of the Cox-weight criterion

The rank-three construction uses four weight spaces, each of dimension three. Their multiplicities are essential to descent. For the full type- I_3 Galois image, increasing the rank separates these weights and prevents the same quotient criterion from applying.

Proposition D.1 (Rank-four subtori and descent). *Suppose that the Galois image on the marked projective Cox model is the full group I_3 . The torus T_0 has a unique rank-four k -subtorus T_4 . Its sixteen Cox-coordinate weights are distinct and have Galois orbits of sizes four and twelve. Consequently no set of five of its weight spaces has a direct sum defined over k , and T_4 cannot satisfy the weight-space descent hypothesis of Theorem 2.1.*

Proof. Write $M = X^*(T_{0,\bar{k}})$, using the character basis $b_1, \dots, b_5$ fixed above (10). A rank-four subtorus corresponds to a saturated, Galois-stable rank-one kernel in M . Each of the two character generators A, B of I_3 must act on such a kernel by a sign. In column coordinates for $b_1, \dots, b_5$, they are

$$A = \begin{pmatrix} -1 & 0 & 0 & -1 & -1 \\ 1 & 1 & 1 & 0 & 2 \\ 0 & 0 & -1 & -1 & -1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ -1 & -1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & -1 & -1 \end{pmatrix}.$$

The stacked matrices

$$\begin{pmatrix} A - \varepsilon I \\ B - \eta I \end{pmatrix}, \quad (\varepsilon, \eta) = (-1, -1), (-1, 1), (1, -1), (1, 1),$$

have ranks $(5, 4, 5, 5)$, respectively. The sole rational line is generated by

$$m = (-1, -1, -3, -3, 3), \quad Am = -m, \quad Bm = m.$$

Its first coordinate is a unit, so m is primitive and the integral kernel is $\mathbf{Z}m$. This proves existence and uniqueness of T_4 .

Restriction of weights is the quotient $M \rightarrow M/\mathbf{Z}m$. In the basis given by the last four standard basis classes, it is

$$(w_1, w_2, w_3, w_4, w_5) \mapsto (w_2 - w_1, w_3 - 3w_1, w_4 - 3w_1, w_5 + 3w_1).$$

The sixteen restricted weights are distinct. Recovering the affine Galois actions, including the common translations of projective weights, gives the orbit

$$\{E_3, E_4, L_{34}, Q\}$$

and its twelve-element complement. The exact calculation of these ranks, weights and permutations is recorded in the rank-four certificate. Since every weight space is one-dimensional, a sum of five weight spaces could descend only if its five weights formed a Galois-stable set. A union of the two orbits has size 0, 4, 12, or 16, never five. $\square$

This failure concerns descent, rather than the integral basis condition. Among all 4,368 five-subsets of the sixteen weights, exactly 1,992 have four differences forming a basis of $M/\mathbf{Z}m$. None is Galois stable. On restriction to T_3 , the twelve-element orbit instead forms the four three-dimensional weight spaces used above. The same orbit argument excludes rank five: the full T_0 weights are already distinct and have the same two orbits, whereas the criterion would require six weight spaces with descended direct sum. Thus rank three is the largest possible rank for this criterion in the full- I_3 Cox representation. The proposition applies to this Cox embedding and the full I_3 action; it makes no assertion about other embeddings, nonlinear sections, or smaller Galois images.

E Exact tangent-section certificate

The four constructions below depend symbolically on the surface parameters (a, b) . For every smooth parameter value, at least one has both the required evaluation rank and a nonzero smoothness minor.

For a tangent point p parametrized by $z = (z_1, z_2, z_3)$, the space H_p of tangent hyperplanes containing B has dimension four. Evaluation on the four selected weight spaces gives a 4-by-4 matrix. The following table records four exact witnesses. In each row, z specifies p , $e = (e_1, \dots, e_5)$ gives the five coordinates $E_1, \dots, E_5$ of x , and the remaining coordinates L_{ij} and Q are reconstructed from e, z' and satisfy all twenty Cox quadrics. The individual values serve only to certify nonemptiness of the required open.

i	z	e	z'
1	(1, 3, 7)	(2, 3, 5, 7, 11)	(2, 4, 9)
2	(2, 5, 11)	(3, 4, 7, 13, 17)	(1, 6, 10)
3	(3, 8, 13)	(5, 7, 11, 17, 19)	(2, 9, 15)
4	(4, 9, 17)	(2, 5, 11, 19, 23)	(3, 10, 18)

Here are the reconstruction conventions. Write $\ell_{ij}(z)$ and $q(z)$ for the following polynomials:

$$\begin{aligned}
\ell_{12} &= z_3, & \ell_{13} &= z_2, & \ell_{14} &= z_2 - z_3, & \ell_{15} &= bz_2 - az_3, \\
\ell_{23} &= z_1, & \ell_{24} &= z_1 - z_3, & \ell_{25} &= bz_1 - z_3, & \ell_{34} &= z_1 - z_2, \\
\ell_{35} &= az_1 - z_2, & & & & & \ell_{45} &= (b-a)z_1 + (1-b)z_2 + (a-1)z_3, \\
q &= b(1-a)z_1z_2 + a(b-1)z_1z_3 + (a-b)z_2z_3.
\end{aligned}$$

Then $p(z)$ has coordinates $E_i = 1$, $L_{ij} = \ell_{ij}(z)$, $Q = q(z)$. For the second point set

$$E_i = e_i, \quad L_{ij} = \frac{\ell_{ij}(z')}{e_i e_j}, \quad Q = \frac{q(z')}{e_1 e_2 e_3 e_4 e_5}.$$

For clarity, the selected Jacobian rows are the derivatives of

$$\begin{aligned}
&E_2 L_{12} - E_3 L_{13} + E_4 L_{14}, & a E_2 L_{12} - b E_3 L_{13} + E_5 L_{15}, \\
&E_1 L_{12} - E_3 L_{23} + E_4 L_{24}, & E_1 L_{12} - b E_3 L_{23} + E_5 L_{25}, \\
&E_1 L_{13} - E_2 L_{23} + E_4 L_{34}, & E_1 L_{13} - a E_2 L_{23} + E_5 L_{35}, \\
&(b-1)E_1 L_{14} + (a-b)E_2 L_{24} + E_5 L_{45}, & a L_{23} L_{45} + (a-b)L_{24} L_{35} - E_1 Q.
\end{aligned}$$

The columns, in order, are $E_1, E_2, E_5, L_{12}, L_{13}, L_{14}, L_{15}, L_{23}$. Evaluate at $p(z)$ to obtain the minor M . These eight Jacobian rows span the tangent-hyperplane space wherever $M \neq 0$. Taking the kernel of their restriction to B gives H_p ; its evaluation on the four selected weight spaces gives D . Thus each asserted open $D_i M_i \neq 0$ supports the construction.

For the first row, the entries in the z column determine the tangent point p , while the entries in the e and z' columns determine x . Linear algebra on the tangent Jacobian, together with the condition that the hyperplanes contain B , gives a basis of H_p . Evaluating that basis on the four weight-space components of x gives the determinant D_1 below. The Jacobian minor M_1 certifies smoothness at p . The other rows repeat this construction with the indicated coordinates.

Let D_i be the determinant of the evaluation matrix for choice i , and let M_i be the fixed 8-by-8 Jacobian minor at the same tangent point. Choice i certifies the evaluation and smoothness conditions on the open $D_i M_i \neq 0$. After discarding nonzero rational factors,

the four evaluation determinants are

$$\begin{aligned}
D_1 &= (8a - 27)^2 (3968a - 1349b - 2619), \\
D_2 &= 568453977a^3 - 373279764a^2b - 1658788404a^2 \\
&\quad + 59068867ab^2 + 976763788ab + 925668016a \\
&\quad - 128537024b^2 - 359163728b - 10185728, \\
D_3 &= 1643918400a^3 - 1105797420a^2b - 7261914240a^2 \\
&\quad + 154718524ab^2 + 3804690316ab + 9944961600a \\
&\quad - 282623593b^2 - 3443769343b - 3454184244, \\
D_4 &= 35068545a^3 + 2864538a^2b - 236376549a^2 \\
&\quad - 5350827ab^2 + 24417420ab + 469658574a \\
&\quad + 14536855b^2 - 90750394b - 214068162.
\end{aligned}$$

and the corresponding smoothness minors are

$$\begin{aligned}
M_1 &= -4(a-1)(b-7)(3a-b-2)(2ab+7a-9b), \\
M_2 &= -9(a-1)(2b-11)(3a-b-2)(4ab+11a-15b), \\
M_3 &= -25(a-1)(3b-13)(2a-b-1)(3ab+13a-16b), \\
M_4 &= -(a-1)(4b-17)(13a-5b-8)(32ab+85a-117b).
\end{aligned}$$

The assertion used in Proposition 3.2 is

$$\{\Delta \neq 0\} \subseteq \bigcup_{i=1}^4 \{D_i M_i \neq 0\}. \quad (31)$$

The opens $D_i M_i \neq 0$ are not contained in $\{\Delta \neq 0\}$; only the containment is claimed.

To prove (31), suppose first that $D_i M_i = 0$ for $i = 1, 2, 3$. Choosing one factor from each product gives eight cases. In the table below, a string such as $D_1 D_2 D_3$ records the three separately vanishing factors, not one product equation. A lexicographic Gröbner calculation in $\mathbf{Q}[a, b, h]/(h\Delta - 1)$ gives

chosen equations	localized zero locus
$D_1 D_2 D_3, D_1 D_2 M_3, D_1 M_2 D_3, D_1 M_2 M_3$	empty
$M_1 D_2 D_3, M_1 D_2 M_3$	empty
$M_1 M_2 D_3$	$3a - b - 2 = Q_0(b) = 0$
$M_1 M_2 M_3$	$3a - b - 2 = (3b - 26)(3b - 13) = 0$

where

$$Q_0(b) = 31223016b^2 - 435944529b + 1306078948.$$

On the line $3a - b - 2 = 0$, the primitive numerator of $D_4 M_4$ is

$$\begin{aligned}
&-(b-1)^4(4b-17)(16b-85)(83246b^2 - 872181b + 2185995), \\
Q_4(b) &= 83246b^2 - 872181b + 2185995.
\end{aligned}$$

The evaluations

$$\begin{aligned}
Q_0(17/4) &= \frac{69121705}{4}, & Q_0(85/16) &= -\frac{4117757269}{32}, \\
Q_4(26/3) &= \frac{7918133}{9}, & Q_4(13/3) &= -\frac{272530}{9}
\end{aligned}$$

exclude the four linear roots. Finally, the two quadratics are coprime by the identity

$$\begin{aligned} & (2265746679974131615 - 377042650728395274b)Q_0(b) \\ & + (141417109727495582904b - 1342668830289072756147)Q_4(b) \\ & = 24176690547344887359179755. \end{aligned}$$

Thus $D_4M_4 \neq 0$ in both remaining cases. This proves that the four opens $D_iM_i \neq 0$ cover $\Delta \neq 0$. The retained certificate gives, for each empty case, an identity expressing a nonzero constant in the corresponding localized ideal.

F Verification and scope

Using exact integer and rational arithmetic, the reconstruction program starts from the two type- I_3 character matrices and the twenty Cox quadrics in [17, Section 3]. With SymPy 1.14.0, it reconstructs the saturated rank-three lattice, residual rank-two torus, Cox weights, tangent Jacobian, four pairs (D_i, M_i) , eight localized cases, and Bézout identity. It also verifies the weight basis, derives the linear section from the kernel of $H_p \rightarrow k$, $\lambda \mapsto \lambda(x)$, and renders every computation-derived value printed in the manuscript into a generated TeX file checked by the verification gate. The certificate retains exact Nullstellensatz identities for the six empty cases, which both verification programs verify directly; the second program also checks the cover, linear section, and quotient minors. The source transcription, quotient criterion, equivariant generic trivialization, descent of the tangent-section conditions, and function-field argument are checked in the text, not by the programs.

For Proposition D.1, a separate certificate exhausts the four sign systems and all five-subsets of the sixteen weights. One program computes determinants in the rank-four quotient; an independent program computes the corresponding rank-five determinants in M , using integer arithmetic without SymPy. It also checks the affine actions and the rank-three weight multiplicities. Negative controls detect an unsaturated character generator and omission of the affine translations.

An additional exact-arithmetic program checks (20)–(23), the seed Jacobian ideal and the ranks 25 and 28 used for the moduli image. For the arithmetic pencil it checks the discriminants, elliptic quotient equations, five j -invariants, and the corrected arithmetic twists against direct point counts. Its finite parameter tests use the bound 240. The symbolic program is a trusted SymPy execution; unlike the Cox and rank-four certificates, its entire calculation has not been reproduced by a second implementation. In particular, the signed Picard identification, Prym comparison, reduction theorem and all-parameter deductions are written arguments and cited inputs, not consequences of sampled point counts. The command and hashes are in the accompanying verification guide.

No Lean development formalizes Theorem 1.1, Theorem 1.3, Theorem 4.1, Theorem 1.2, Theorem 2.1, Proposition 3.2, or the stated corollaries. The formal-coverage annotations therefore say **absent**.

We import Tschinkel–Zhang’s universal-torsor geometry and their classification of the four Galois images for which H^1 vanishes on every subgroup. We construct the rational quotient by a rank-three subtorus and its descended tangent section. This construction lowers the stabilization bound to two; no classification beyond those four Galois types is needed here.

AI assistance disclosure

This project was developed with extensive assistance from OpenAI Codex and Anthropic Claude. Under the author's direction, the systems assisted with proof exploration, literature research, exact computation, verification, and manuscript drafting and revision. The author checked the resulting arguments, computations, code, and cited sources, reviewed and edited all AI-assisted material, and assumes responsibility for all content.

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